

AE5050 Internship
Title page internship report

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Preface

This internship was provided by SRON Netherlands Institute for Space Research. It was completed in the Earth Science group, from June 15th to October 15th. Being in close relation with the urgent global challenge that is climate change, the topics of the work and research done throughout this internship, aligned with and only reinforced my past interests to contribute to solving this problem. Indeed, the research on anthropogenic greenhouse gas emissions done within institutes like SRON is essential to tackling this issue, by providing accurate estimates of the former and context of the current environmental situation to larger bodies and governments to ensure the effectiveness of current regulations and assess our progress towards a greener future. Thus, with the major, potentially devastating implications that come with climate change, at our doorstep, our generation must face those challenges head on, from today onward, and adopt a more sustainable way of living.

I would like to thank my supervisor for this 4 month period, Friedemann Reum, who has been very supportive and available throughout this research process, despite the less than ideal work conditions imposed by the global pandemic. I would also like to thank SRON and the Earth program for this opportunity, which broadened my professional experience and brought me great insight into the world of research.

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Abstract

Accurately estimating terrestrial carbon inverse fluxes depends on an array of parameters, one of the most important being atmospheric transport model errors. This research attempts to quantify the carbon dioxide column mole fraction uncertainties that arise from these transport model errors, specifically targeting the CO₂ emissions plume of Berlin, using the Weather Research and Forecasting (WRF) model, over Eastern Germany during winter. This goal is achieved by first studying the WRF model's physical parameterization options and past scientific literature that provides strong foundations on the topic. Then, an ensemble of models with varying physics configurations is created. These simulations are run with different planetary boundary layer (PBL) schemes, land surface models (LSMs), cumulus parameterizations (CPs) and urban surface (US) schemes. The rest of the physical parameters are fixed, along with the meteorological reanalysis datasets, ERA5, and the CO₂ boundary conditions datasets, from the Copernicus Atmosphere Monitoring Service (CAMS). The simulated CO₂ city plume signal is then precisely characterized, to perform statistical analysis between pairs of models differing in only one physical parameterization and quantify the uncertainties obtained. The latter step provides a strong representation of the relative contribution of each physics parameterization to the uncertainty in CO₂ column mole fraction estimates. Finally, the contributions of other simulated signals, from the background, biosphere and other anthropogenic sources (such as power plants) to the uncertainty of CO₂ column mole fraction estimates over Berlin, are also determined. For all simulated signals, the mean signal size uncertainty is represented by the Mean Absolute Difference (MAD) between pairs of models that differ in one physical parameterization. Varying the PBL scheme produces the largest city plume signal uncertainty out of the tested physical parameters, with a MAD of 0.077 ppm, for a mean city plume signal size of 1 ppm. The LSM has a slightly smaller, yet comparable effect, with a MAD of 0.07 ppm, followed by the US scheme, with a MAD of 0.04 ppm. The CP is found to have almost no effect on the uncertainty of the simulated CO₂ signals. The relative contributions of PBL scheme, LSM and US scheme are similar in the other simulated signals. They also generally agree with existing scientific literature on the topic, while that of CP is particular to the high resolution research setup explored in this study. It is also concluded that the signal formed by other anthropogenic sources is subject to the largest uncertainties in simulated CO₂ emissions, going up to 15% of the mean city plume signal size, followed by uncertainties produced by the city plume signal itself. Whereas the biosphere and background signals, although larger in magnitude, exhibit very small estimate uncertainties. The obtained results also show that, while atmospheric data is another important contributor to inverse carbon flux estimate uncertainties, and must become more accurate and spatially complete, the modeling community also has some margin of improvement regarding transport model errors, as resulting uncertainties in CO₂ column mole fractions aren't negligible.

1 Report Introduction

This report describes the detailed research process that took place throughout this 4 month period. The supervisor for this internship has been working on CO₂ emissions simulations over the city of Berlin, using the WRF model and was interested in exploring the uncertainties of modeled total atmospheric column CO₂ mixing ratios as seen by satellites such as the future CO2M mission. While such satellite missions provide more accurate and complete atmospheric measurements, thus reducing the related uncertainties, the research on the modeling of atmospheric transport is also essential in better estimating carbon fluxes, and emission prediction. Thus, the research for this internship focuses on studying different WRF configurations of the physics options for higher domain resolutions, along with quantifying the uncertainty of resulting simulated CO₂ mixing ratios. Some of the problematics covered are: Which WRF physics and dynamics configurations are suitable for high-resolution CO₂ transport modeling? What is the impact of the configurations on modeled total column CO₂ mixing ratios? What is the respective contribution of background and plume mixing ratios to the uncertainty?

The report answers these questions using the following detailed structure. Firstly, Chapter 2 gives an overview of SRON, its role in the scientific community, and its main research topics. In Chapter 3, the problem which this internship attempts to solve, is introduced, within its broader context, along with the steps taken to complete the project. Then, Chapter 4 presents the work done to answer the research questions and the methodology used. Chapter 5 holds the results and consequent discussion regarding the quantification of WRF uncertainties. Finally, the conclusions in Chapter 6 provide answers to the posed problematics and finalize the study. Some recommendations for future related work are also suggested in Chapter 7, to orient the focus towards other facets of the problem at hand.

Although no extensive technical background is required to understand the report, here is a brief introduction to the WRF model, used throughout this internship. The Weather Research and Forecasting (WRF) model is a widely used weather prediction numerical model. Its applications mostly fall into two categories: operational weather forecasting (news, phone apps), and research on the atmosphere [2]. As most numerical models, it takes meteorological initial and boundary conditions as inputs, and propagates the state of the mesh through time, using modeled physical and dynamic processes. Thus, a range of parameters and schemes are used to simulate accurately surface and atmospheric exchanges and transport. The WRF model can also introduce gases and chemical processes within the modeled environment, allowing researchers to study the behavior of greenhouse gases in the atmosphere and related fluxes with the surface.

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3 Problem Introduction

3.1 Overarching Context

In the last 60 years, a steady increase in temperatures on Earth has taken place, more commonly referred to as global warming. The consequences of this climate change, which have repercussions of their own, include sea level rise, an increase in extreme weather events, risks for wildlife, human health, agriculture, and more [5]. The physical process at the root of this evolution is the enhanced greenhouse effect, which describes the increased heat retained within Earth's atmosphere, due to rising concentrations of trace gases such as carbon dioxide and methane. Indeed, data from the paleo-record, over hundreds of thousands of years, deduced from ice cores has shown that surface temperatures are undoubtedly linked to CO₂ levels in the atmosphere [6]. The two main culprits causing this increase in trace gases have been identified as land use change and fossil fuel burning. Indeed, as the biosphere plays a key role in the carbon cycle, absorbing an important portion of carbon dioxide emissions, deforestation has a direct impact on the equilibrium of the system. Then, the large increase in fossil fuel burning since the second industrial revolution, at the beginning of the twentieth century, only inflates this environmental imbalance. Although the global picture of climate change, described above is well known in the scientific community, its processes and subtleties at the "policy-relevant" scales are still rather inaccurately represented. Thus, further research on the spatial and temporal distribution, along with the precise magnitude of terrestrial carbon fluxes is imperative to better predict the evolution and behavior of these trace gases, for precise emission reduction regulations to be set and followed.

3.2 Problem Statement

There exists two methods to estimate these carbon fluxes: the bottom-up and top-down methods. The first consists in modelling local biogeochemical processes at the surface, and using data, maps and specific variables from biomass in a given ecosystem to estimate the fluxes. The second method consists in simulating atmospheric variations in trace gases to then temporally and spatially estimate the sources and sinks steering the system [7]. The latter is the method used throughout this internship (specifically for carbon dioxide) and the only one referred to throughout the rest of the report. Its goal is to reach a perfect match between simulated CO₂ concentrations and observed CO₂ data, for the flux estimates to be optimal. However, as with any complex numerical model, the inverse method falls victim to uncertainties and errors. The latter originate from different sources, such as incomplete atmospheric data, but also model-related ones, like transport errors and grid resolution limitations [8]. While, more accurate monitoring instruments are being developed and equipped on orbiting satellites to better the atmospheric data sets required, it is also the modeling community's duty to improve inverse models for more reliable flux estimations. Thus is the overarching reason for this internship and research.

3.3 Research Objectives

Within the research domain of transport model errors, exists many different areas that ought to be studied. However, time restrictions in such projects only allow for specific topics to be explored in depth. Thus, the main goal of this internship is to quantify uncertainties in CO₂ column mole fractions due to atmospheric transport errors in the WRF model, and assess the contribution of the different atmospheric carbon dioxide signals to said uncertainty. Indeed, the total CO₂ concentrations visible in the atmosphere are the result of different sources, namely the biosphere, anthropogenic emissions (from large cities, power plants, airports and such human led enterprises) and the background (natural concentrations). Thus, it is of interest to determine the individual contribution of each simulated signal, to the total uncertainty in WRF. Increasingly more research explores these topics (Diaz-Isaac et al., 2018; Sarmiento et al., 2017), attempting to determine optimal WRF parameters to minimize said uncertainty.

However, these studies generally remain inconclusive regarding the "best" WRF options, due to their intrinsic differences in region, settings, and other simulation elements. Every parameterization option in WRF leads to differences in the results, meaning that even slight mismatches between two setups will inevitably lead to inconsistent conclusions. Indeed, for example, a difference in input datasets in one study, could partly cause a positive bias in the PBL variables (PBL height, wind speeds and directions) and in turn the CO₂ mole fractions, although the same main physics schemes are used. Thus, the vast number of variable parameters in WRF still gives way for more related research. This study then focuses on the uncertainty obtained using the most popular and widely used WRF schemes, in the context of high resolution modeling, following the research led by this internship's supervisor, F. Reum, which looks mainly at the city plume of Berlin.

3.4 Project Structure

The first step of the project consists in familiarizing oneself with the general setting and context of the research. Then, understanding how the WRF model actually works, and knowing its constituents, is necessary to work successfully with the data sets and results. The WRF model is run on the Dutch national supercomputer Cartesius, offering the large storage space and computing power needed to run the former. Thus, getting acclimated to this UNIX based system is also part of this first phase. The second phase consists of precisely defining the domain, to include the city plume of Berlin, studied throughout the rest of the project, basing the work on the material provided by F. Reum. Then, the next part is to fully run the WRF model, which is achieved with python and shell scripts to process the data, run the model and extract the relevant results. Finally, the third and last phase consists in actually answering the research problematics. An ensemble of models with varying WRF configurations is developed, in order to obtain a range of results, illustrating the uncertainty in CO₂ mole fractions studied. The ensemble definition for this project is based on the domain characteristics (e.g. high resolution, city plume), and a review of the literature that revolves around parameterization sensitivity studies and physics options testing in the WRF model. This work thus includes different PBL schemes, LSMs, CPs, and US schemes. A set of metrics is then defined to assess and quantify this uncertainty, along with a rigorous comparison method from model to model, to accurately analyse and interpret it. Finally, the research questions are answered and some insightful conclusions are drawn from the overall project. It is important to mention that this type of research is bound by time restrictions, which only allow one to explore certain aspects of a project, and not others. Indeed, in this case, it would be interesting to utilize other analysis metrics, and test more physical and non-physical parameterizations. Furthermore, the limited time also didn't allow for studying the effect of these transport error uncertainties on CO₂ flux estimates.

4 Methodology and Work Done

This section of the report covers the work done throughout the internship, to answer the research questions posed earlier. It describes the tasks performed, resultant products and personal growth, while also explaining some contextual pieces of information, when relevant and required to understand the rest. As this internship is research based and encompasses one large project, rather than multiple smaller ones (often the case in non-research based companies), the subsections are ordered chronologically. Indeed, the research familiarization process is presented first, followed by the domain definition. Then, the work done to run the WRF model is described, and a first look at a simulation is shown. Furthermore, the ensemble definition is presented, through the literature study done and reasoning behind the configuration choices. Finally, the metrics and methodology developed to analyse the results are described and justified.

4.1 Research Familiarization

The first step to the project consists in educating oneself on the concepts related to the research being done. Thus, the following subsections present an overall review of the topics relevant to the project, namely atmospheric CO₂ transport and the WRF model itself.

4.1.1 Atmospheric Transport

Atmospheric transport refers to the circulation processes within Earth's atmosphere, governed by gravity, pressure gradients and the Coriolis effect [9]. To have a reference of understanding for further explanation, the fluid mechanics analogy of a fluid near a plate is used as follows. When a fluid comes in proximity to a surface, a plate for example, its flow is disrupted by larger viscous forces. The layer of fluid which is affected is called the boundary layer, within which its velocity varies, from nil at the plate surface to its regular velocity at the boundary layer limit. The flow may be laminar (orderly and smooth) for slow, viscous fluids, or turbulent, for fast, low viscosity fluids on a rough surface. A turbulent flow in the boundary layer exhibits strong vertical mixing and large flow fluctuations. Similarly, Earth's atmosphere has a planetary boundary layer (PBL) from the surface of the Earth to 1 or 2 kilometers in the atmosphere. Within it, the air flow is turbulent, with rapid changes in temperatures, velocities and mixing ratios, of trace gases for example. Above it, the air flow is free of any surface viscous effect, and thus geostrophic parallel to isobars [10]. A few challenges arise when trying to model atmospheric transport within this PBL. Firstly, as a numerical model, the WRF model represents the surface as a grid of points, of which the resolution cannot exceed a certain limit, due to computing power constraints. Furthermore, the atmospheric data used for initial and boundary conditions, as well as model adjusting, is often spatially incomplete, which affects the accuracy of simulations. What's more, the meteorology and transport are also affected by elements like Earth's surface roughness, which varies from cities to rural or inhabited areas. Indeed, the more buildings and "uneven terrain", the larger the affect on wind velocities, as seen in Figure 1. Finally, as turbulence is a stochastic process, it cannot be modeled explicitly. Thus, the WRF model and others, parameterize sub-grid scale processes such as turbulent mixing, through a combination of physics schemes and options to optimize the transport accuracy. These schemes model elements like the PBL, cloud formation and behavior, microphysics, the land surface, while additional parameters include options like resolution, and nudging (used to tip the model mid-simulation towards observations). Hence, in order to study CO₂ mixing ratio uncertainties due to transport errors, one must look at the way WRF parameterizes transport, and consequently at the schemes and options.

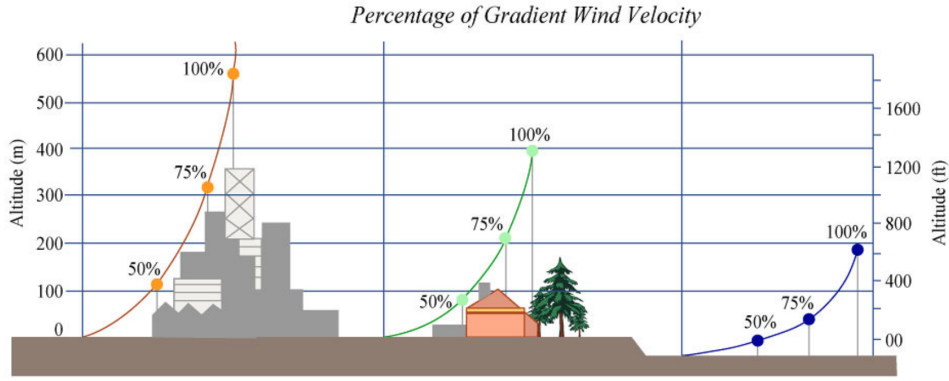


Figure 1: Effect of terrain on wind velocity Profiles. [1]

4.1.2 WRF Model Structure and Functions

It is also essential to understand the WRF model structure and components, as it doesn't run straight by itself. There are two main steps to run the complete model: the WRF Pre-processing System (WPS) and the WRF model. The WPS consists of three programs which prepare the domain and meteorological data necessary to run the model, using a personalized WRF file named 'namelist.wps' containing initialization information. Indeed, the WPS first sets the domain grid and introduces the required terrestrial data to the grid. It then converts the necessary meteorological fields data from other models into a readable format, to finally fit it to the defined grid [11]. Once this process is achieved, the second step of running the WRF model can take place, using another personalized WRF file named 'namelist.input', which contains more information on the domain, the duration and parameterizations. This step begins with a program to create the initial and boundary conditions from provided datasets. Then, the last program integrates the model and produces the meteorological simulation desired, with modeling climate variables like wind speeds, rainfall, albedo, and more [12]. In the case of atmospheric gases research, the WRF model is coupled to a chemistry option which introduces gas transport in the simulations. In the case of this internship, CO₂ emissions datasets are fed into and integrated by the model. The resulting simulations can then also show atmospheric CO₂ mixing ratios, and the different constituent signals. The following sections present the work done during the internship to run the model as presented above.

4.2 Domain Definition

The region studied during the internship covers Eastern Germany, and more precisely Berlin and some neighboring power plants. The reason for this choice is rooted in the nature of the supervisor's research, which focuses on this area. Although the domains aren't exactly identical between both studies, this choice made the pre-processing step easier, as data sets were similar. Furthermore, this provided a base of reference for this internship's study, where initial results could be validated by comparing the simulations. This was also ultimately the main reason for this internship: to further the understanding of varying the WRF parameterization options and the effects on simulated CO₂ mixing ratios from Berlin's city plume. What's more, for this region are available well-founded datasets for CO₂ emissions from the Netherlands Organisation for Applied Scientific Research, TNO, as well as for boundary conditions of natural emissions from the CAMS.

The specific characteristics selected for the domain are as follows. Firstly, most WRF simulations make use of nesting, which consists in running the model with a higher resolution nested domain within a coarser resolution parent domain. This method is used to avoid running models at high resolutions over large domains, which is too computationally expensive. Furthermore, the parent domain can provide

more spatially and temporally adequate lateral boundary conditions to the nested domain than ones available without the parent domain [13]. Thus, a nested domain, d02, with a 2km by 2km resolution is defined, within a parent domain, d01, with a resolution of 10km by 10km (see Figure 2). The nested domain d02, covering Berlin and some neighboring power plants, is the domain studied throughout the internship. A few reasons justify the chosen domain resolutions. Other than being a good compromise between high resolution and computational cost relative to the d02 domain size, the 2km by 2km resolution corresponds to the footprint of a single sounding of the Copernicus CO₂ Monitoring Mission (CO2M) satellite constellation planned by the EU [14]. Running WRF on the same resolution, one pixel of the simulation can be directly compared to what the satellite records. Although the footprints won't exactly align with the model grid, this could be an interesting future aspect of the research. Another factor in the choice of resolutions is the cumulus parameterization gray zone. Indeed, for resolutions larger than 10km, the cumulus processes are parameterized, while at scales below 3 to 4km, the model is assumed to solve organized convection by itself [15]. However, questions regarding the requirements needed in this in-between resolution zone remain, hence the choice to avoid it all together in this study. Then, there are 40 vertical layers defined, for both domains, representing the atmospheric column. Finally, the models are run over a period of 6 days, from February 9 to February 14. This time frame constrains the simulation duration to a few hours, and is also long enough for the CO₂ plumes to fully develop and evolve. The dates also correspond to F. Reum's simulation setup, making it possible to verify early results.

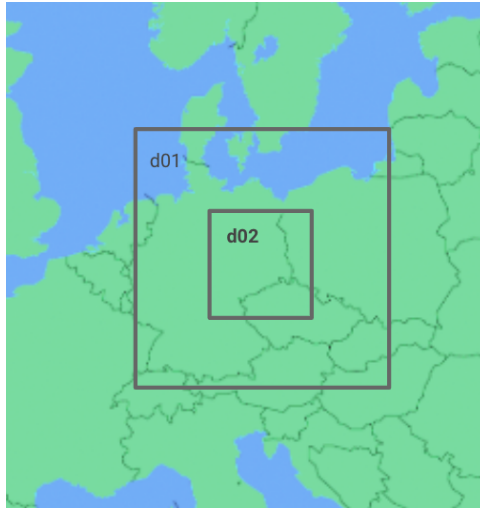


Figure 2: Geographical domains used for study. The inner domain, d02, is where the research is focused.

4.3 Running the WRF Model

As previously mentioned, other than computing power and large storage space both provided by the Dutch national super computer Cartesius, running WRF requires some preparation, in the form of input files, scripts to format data and execute the run. This work is based on material provided by the supervisor, F. Reum, which is adjusted to fit the study at hand. The first phase in running the model, is to run the pre-processing system WPS. Then the goal is to run WRF, and obtain a basic meteorology simulation, before finally introducing the CO₂ emissions to the domain.

Successfully running WPS starts with personalizing the standard namelist.wps file which the model needs to define the domains, grids and fit the meteorological fields to the latter. Thus, the file is fitted for a simulation with 2 domains, from February 9 to February 14. The domains are precisely characterized, with their spatial and temporal resolutions, number of pixels, and their center's longitude and latitude, 13.5° and 51.5° respectively. The map projection used is the Lambert conformal conic projection. Then, a personal directory on the Cartesius super computer's UNIX based system is created. It is then organized and set up to store the input meteorology datasets. The atmospheric data used is ERA5 from the

Copernicus Climate Change Service. Further along the process, a run script in the Python coding language, from F. Reum, is used to indicate to the model the location of the required datasets and files, within the UNIX based directories. The Anaconda program is used to facilitate the management and use of python packages and modules on Cartesius. To not have to manually modify the run script for every different simulation, files of simulation-dependent settings are used, of which the variables are called in the run script. These settings files thus mostly contain directory locations for files and data, in a .txt format. To more easily transfer and utilize these files, they are serialized into byte streams. This step is achieved by creating a Python script that converts the file format, using the pickle module. The necessary information is hence transferred through these small personalizable settings files, while the large python run script remains common for all simulations. Thus, the first simulation results from WPS can be visualized using NASA’s plotting software for georeferenced data, Panoply. Figure 3a shows wind velocity fields over the nested domain, obtained with WPS.

Once WPS works correctly, one can focus on running the rest of the model. First, the second personalizable standard file, named namelist.input, is completed. It contains options regarding the physical parameterizations, dynamics, nudging, and other more advanced settings, used for the model. While later in the study, the physics schemes are varied to study the resulting uncertainties, the former, along with the rest of these model options are at the moment kept unchanged as this step only consists in running the model successfully. Then, boundary conditions from CAMS data are stored on Cartesius, as well as anthropogenic CO₂ emissions from TNO’s inventory for Europe. The settings file is adjusted accordingly, with the locations of the new directories and files. From there, a debugging work takes place. Indeed, it is common for WRF to require some adjustments before running. Thus, WRF’s error files are used to locate the errors present in the run process. However, as these files do not always explicitly state the problem, this becomes a slightly tedious part of the work. With adjustments to the Anaconda environment, python modules and run script used, as well as the settings and namelists files, the task of running WRF can take longer than expected. Once operational, the WRF model can simulate CO₂ column mole fractions over the nested domain, from sources such as the city of Berlin, the biosphere, background and other anthropogenic sources like power plants. Additional shell scripts are used to extract only the results relevant to the study from the large WRF output files. New variables are then created, where the background emissions are subtracted to better visualize the different CO₂ signals. The CO₂ mixing ratios are also vertically averaged to simplify the rest of the study, to ultimately not exceed time limitations. An example of Berlin’s averaged CO₂ city plume signal is shown in Figure 3b, where individual point sources can be identified.

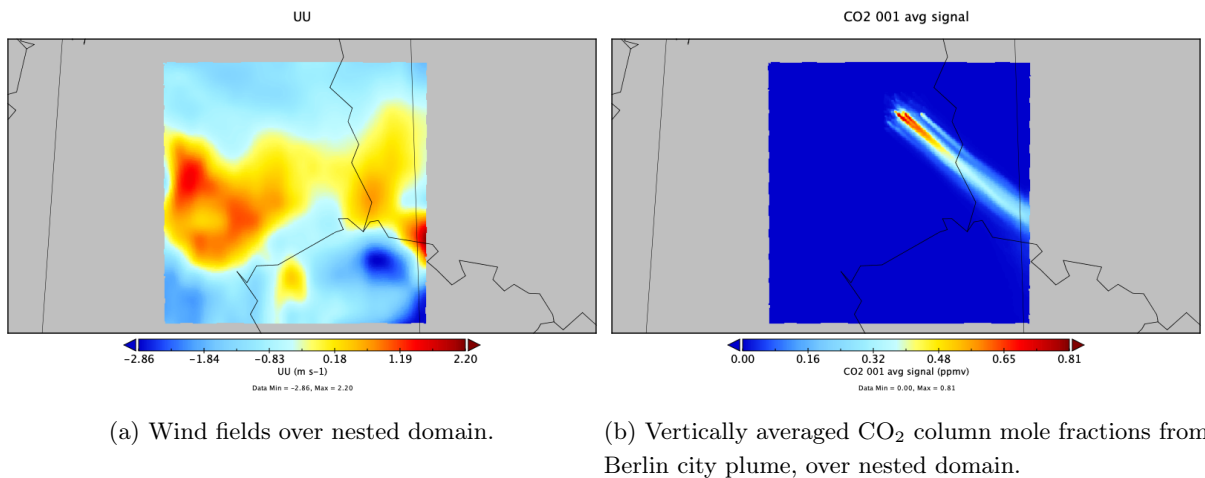


Figure 3: First simulation results with WRF.

4.4 Literature Study and Ensemble Definition

After having setup the global environment necessary to run the WRF model, it is now possible to focus the project on answering the research questions posed. Thus, to quantify the carbon dioxide column mole fraction uncertainties that arise from transport model errors in WRF, targeting the city plume of Berlin, an ensemble of models that differ in physical parameterizations is assembled. The physics selected to be varied are ones with the largest effect on the transport uncertainty, and with highlighted relevance to the study, while the schemes' selection criteria are their reported accuracy and also relevance. This evaluation is based on the ongoing research by F. Reum, and a literature review of similar sensitivity studies with WRF. Finally, the ensemble is presented along with its required adjustments.

It is important to note that, although a literature review provides a strong foundation in knowledge and previous work on the topic, these scientific papers should not be taken as absolute truths for future work in the field. Firstly, every study defines its own specific domain, which in turn affects the selection of other parameters, such as the datasets and resolution used, thus most likely differing from one's own. Although this might seem obvious, when reviewing a paper and analysing in depth the results and discussion, one can easily forget that both projects fundamentally differ in scope and global setup. Furthermore, most WRF parameterization sensitivity studies also differ in the variables assessed (CO₂ mole fraction, temperatures, precipitation, surface energy fluxes, etc.), which implies that one specific WRF physics option might have a larger effect on one variable, than another. Finally, most studies on the topic do not justify the choice of schemes made for the sensitivity analysis, which in turn makes it difficult to select the most reliable ones for this study. Hence, a measure used to make this selection possible is a scheme's "credibility", or number of times it is referenced in the reviewed literature. Thus, it is critical in this step of such a project to review multiple related sources and make a selection accordingly, to remember the scope and goal of one's own project, and also keep in mind that no single option is in every situation the right option.

4.4.1 Planetary Boundary Layer (PBL) Schemes

The first physical parameterization to be varied in the ensemble of models is the PBL scheme. Intuitively, as the PBL's turbulent flow drives rapid vertical changes in moisture, temperature, wind speeds and other boundary layer variables, consequently playing an important role in atmospheric transport and thus transport error when simulated, the variation in PBL schemes should generate relatively large uncertainties in resulting CO₂ mixing ratios. This train of thought is supported by results obtained in sensitivity studies. Other than being the most selected option in similar sensitivity studies (Diaz-Isaac et al., 2018; Feng et al., 2016; Lucas et al., 2017; Nehrkorn et al., 2013; Sarmiento et al., 2017), the PBL parameterization is found to have one of the most substantial impacts on simulated atmospheric CO₂ transport from evaluating its root mean square difference (RMSD) in simulated CO₂ column mole fractions [8]. It also has been shown to greatly affect near-surface winds, which can play a significant role in plume transport uncertainty [16] [17]. The PBL schemes used are the Mellor–Yamada Nakanishi Niino (MYNN) Level 2.5 scheme (coupled to its corresponding MYNN surface layer scheme) and the Yonsei University (YSU) scheme (coupled to its corresponding MM5 Similarity surface layer scheme), two of the most widely used schemes across WRF research. The MYNN has been found in some cases to produce the most accurate PBL variables which translates to simulated transport reliability [18], as well as the most reasonable overall results [8]. Meanwhile, the only reviewed study that shares the region of interest of Berlin with this project, uses the YSU scheme, found to generate smaller biases in modeled near-surface PBL variables than another popular PBL scheme, the Mellor–Yamada–Janjic (MYJ) scheme [19]. Furthermore, the YSU scheme is also the current scheme used by F. Reum, for Berlin city plume simulations.

4.4.2 Land Surface Models (LSMs)

The second physical parameterization varied in the ensemble is the LSM. The latter uses driving datasets of surface variables to simulate terrestrial fluxes, which directly affect transport within the PBL, and thus CO₂ mixing ratios. The LSM parameterization is found to have the largest impact on simulated atmospheric CO₂ transport from evaluating its root mean square difference (RMSD) in simulated CO₂ column mole fractions [8]. Similarly to the PBL scheme, it has been shown to greatly affect near-surface winds, which can play a significant role in plume transport uncertainty [16] [17]. Another study that focuses specifically on LSMs concludes that the latter strongly influence modeled temperatures, which affect transport [20]. The LSMs used for the study are the Noah land surface model and the NCAR Community Land Model version 4 (CLM4). The Noah LSM is the most widely used scheme among the community (Deng et al., 2017; Diaz-Isaac et al., 2018, 2019; Feng et al., 2016; Jin et al., 2010; Kuik et al., 2016; Lauvaux and Davis, 2014; Lucas et al., 2017; Nehrkorn et al., 2013; Sarmiento et al., 2017; Sharma et al., 2017), due to its versatility and mid-level complexity. It is also used by F. Reum, which makes this scheme relevant to the current research context. The CLM4 scheme is the most sophisticated LSM, which renders it rather computationally expensive. However, its predecessor CLM3 was found to greatly improve simulated temperatures compared to other LSMs, like Noah and the Rapid Update Cycle (RUC) model [20].

4.4.3 Cumulus Parameterizations (CPs)

The third physical parameterization to be varied in the ensemble of models is the CP. This physics option simulates unresolved convection in the PBL. It is found to have a similarly substantial impact than the PBL scheme, on simulated atmospheric CO₂ transport from evaluating its root mean square difference (RMSD) in simulated CO₂ column mole fractions [8]. As mentioned in Section 4.2, domains with higher resolutions (less than 3-4 km), like d02 here, resolve organized convection by themselves without a CP [15]. Thus, only the parent domain in this study has a CP. The CPs used in the ensemble are the Grell 3D scheme and Kain-Fritsch (KF) scheme. The Grell-3D scheme is used in multiple other studies as well (Feng et al., 2016; Diaz-Isaac et al., 2018, 2019) and is an improved version of the Grell-Devenyi (GD) scheme, itself used in Nehrkorn et al., 2013. What's more, this scheme is also well-suited for higher resolution domains, making it relevant to this study [8]. The KF scheme is also widely used in sensitivity studies (Deng et al., 2017; Diaz-Isaac et al., 2018; Sarmiento et al., 2017; Sharma et al., 2017). It has been found in some research to be the best suited for winter precipitation, which fits in this study, along with producing the best simulated temperatures, among other CPs [21].

4.4.4 Urban Surface (US) Schemes

The fourth and last physical parameterization varied in this ensemble is the US scheme. This physics option is used to better represent the spatial structure of urban areas and the consequent modified behavior of boundary layer transport, which depends on surface characteristics like roughness, as explained in Section 4.1.1. It is often omitted in WRF sensitivity studies as it only really affects simulations over cities, which aren't always the study subject. However, with the research focusing on the city plume of Berlin, the US scheme is very relevant. Indeed, it has been shown that a clear improvement in near-surface PBL variables over an urban area results from using such a scheme [22]. Thus, the ensemble has models with a US scheme, and others without, to assess the variability in resultant CO₂ distributions over a city due to this parameterization. From the three available US schemes, the one used is the Single Layer Urban Canopy Model (SLUCM). Past research comparing single and multi-layer models has shown no particular advantage to using the latter, more complex and computationally expensive scheme, less compatible with other physical parameterizations [23] [24] [25]. Urban parameters for Berlin (building heights, street widths, etc.) are taken from Kuik et al. (2016), where the city layout is precisely studied [19].

4.4.5 Model Ensemble

The first version of the ensemble (see Table 1) is thus comprised of 16 configurations, which differ in PBL schemes, LSMs, CUs and US schemes. All models share the same remaining settings and options. The microphysics scheme used is the WRF Single-Moment 5-class (WSM5) scheme, which is widely used in the reviewed literature (Deng et al., 2017; Diaz-Isaac et al., 2018, 2019; Feng et al., 2016; Sarmiento et al., 2017), and also in the current research by F. Reum. The short-wave and long-wave radiation scheme used is the RRTMG scheme, which itself is the Rapid Radiative Transfer Model (RRTM), very widely used in similar studies, but modified for General Circulation Models (GMCs) [26]. This scheme, also used by F. Reum, is thus well-suited for simulating atmospheric changes due to the enhanced greenhouse effect.

Model Number	PBL scheme	LSM	Cumulus Parameterization	Urban Surface scheme
1	MYNN (5)	Noah (2)	Grell-3D (5)	None (0)
2	MYNN (5)	Noah (2)	Kain-Fritsch (1)	None (0)
3	MYNN (5)	CLM4 (5)	Grell-3D (5)	None (0)
4	MYNN (5)	CLM4 (5)	Kain-Fritsch (1)	None (0)
5	YSU (1)	Noah (2)	Grell-3D (5)	None (0)
6	YSU (1)	Noah (2)	Kain-Fritsch (1)	None (0)
7	YSU (1)	CLM4 (5)	Grell-3D (5)	None (0)
8	YSU (1)	CLM4 (5)	Kain-Fritsch (1)	None (0)
9	MYNN (5)	Noah (2)	Grell-3D (5)	SLUCM (1)
10	MYNN (5)	Noah (2)	Kain-Fritsch (1)	SLUCM (1)
11	MYNN (5)	CLM4 (5)	Grell-3D (5)	SLUCM (1)
12	MYNN (5)	CLM4 (5)	Kain-Fritsch (1)	SLUCM (1)
13	YSU (1)	Noah (2)	Grell-3D (5)	SLUCM (1)
14	YSU (1)	Noah (2)	Kain-Fritsch (1)	SLUCM (1)
15	YSU (1)	CLM4 (5)	Grell-3D (5)	SLUCM (1)
16	YSU (1)	CLM4 (5)	Kain-Fritsch (1)	SLUCM (1)

Table 1: Initial version of ensemble: 16 models based on literature review. The schemes’ abbreviations are used, followed by their corresponding identification number in WRF options.

Before moving on to analysing the results from the ensemble, all models must be run to test for any unforeseen issues. The first problem encountered relates to the KF cumulus parameterization; all models using this scheme fail to run. This unexpected behavior is however mentioned in the WRF User Guide. Indeed, if cumulus parameterization is required, only the Grell-3D and Grell-Freitas schemes can be used without having to change other settings, which worsen simulation results [27]. Thus, the KF scheme is replaced throughout the ensemble with the Grell-Freitas scheme. This should however not negatively impact the robustness of the ensemble definition, as recent studies have shown that the Grell-Freitas scheme can lead to steadier atmospheric dynamics [8]. The second and last ensemble configurations related issue has to do with a scheme incompatibility. Indeed, models with both the high complexity CLM4 land surface model and the parameterized SLUCM also crash. This is simply due to an incompatibility between those two schemes, which is indicated in the resultant error files. Thus, those 4 models are removed from the ensemble. The new and final version of the ensemble (see Table 2) is set, with all models running successfully.

Model Number	PBL scheme	LSM	Cumulus Parameterization	Urban Surface scheme
1	MYNN (5)	Noah (2)	Grell-3D (5)	None (0)
2	MYNN (5)	Noah (2)	Grell-Freitas (3)	None (0)
3	MYNN (5)	CLM4 (5)	Grell-3D (5)	None (0)
4	MYNN (5)	CLM4 (5)	Grell-Freitas (3)	None (0)
5	YSU (1)	Noah (2)	Grell-3D (5)	None (0)
6	YSU (1)	Noah (2)	Grell-Freitas (3)	None (0)
7	YSU (1)	CLM4 (5)	Grell-3D (5)	None (0)
8	YSU (1)	CLM4 (5)	Grell-Freitas (3)	None (0)
9	MYNN (5)	Noah (2)	Grell-3D (5)	SLUCM (1)
10	MYNN (5)	Noah (2)	Grell-Freitas (3)	SLUCM (1)
11	YSU (1)	Noah (2)	Grell-3D (5)	SLUCM (1)
12	YSU (1)	Noah (2)	Grell-Freitas (3)	SLUCM (1)

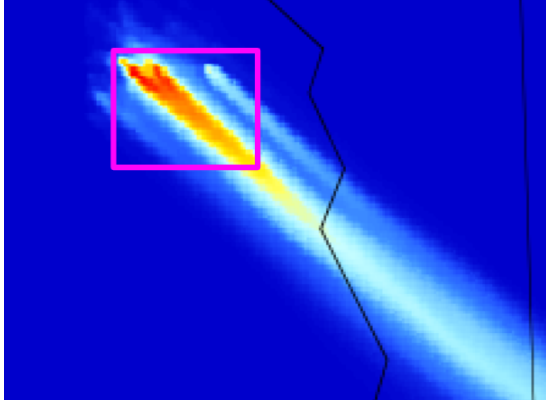
Table 2: Final version of ensemble: 12 models based on revision. The schemes’ abbreviations are used, followed by their corresponding identification number in WRF options.

4.5 Analysis Metrics and Methodology

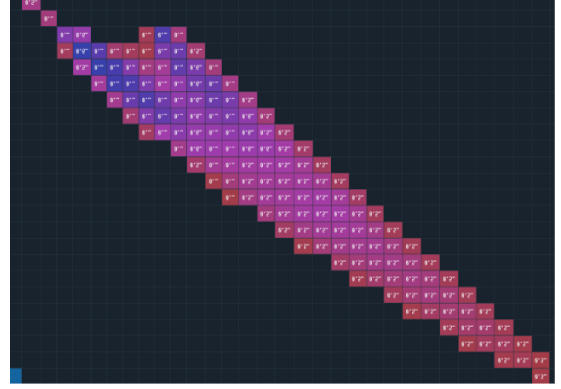
The goals of this project are to quantify the carbon dioxide column mole fraction uncertainties that arise from WRF transport model errors, and evaluate the contribution of different signals to these uncertainties, with respect to Berlin’s city plume. In order to answer these problematics, it is first necessary to define the meaning of a plume. Then, metrics to characterize the plume and signals are needed, to later be able to analyse the differences in models’ results. Finally, one must establish a method to effectively and systematically compare the simulation results, in order to ultimately quantify the uncertainties present, and signal contributions. A python script is then produced to processes the desired WRF output data, apply the defined analysis tools and concepts, and create resultant plots. Is is also important to note that the analysis only focuses on results from February 12 to February 14, as WRF has a certain spin-up time only after which the results normalize and become reliable.

4.5.1 Plume Definition

An emission plume is a streak of trace gas that emanates from a precise source, such as a city or an airport, and propagates in the atmosphere. Such emissions are however of small magnitude compared to natural fluxes, which makes them hard to detect and define. After simulating transport, it is possible to subtract those large natural fluxes from the overall picture, to better visualize the smaller plumes and sources, as depicted in Figure 3b. When examining such data in detail, it is necessary to assess it pixel by pixel, to perform a quantitative and approachable analysis. Thus, the plume at a time t is defined by every pixel for which the variable of interest, here CO_2 mole fraction, is above a certain threshold. No categorical value exists for this threshold, as this analysis definition process is relatively dependent on the study at hand. Thus, to remain in the scope of potentially comparing results to the CO2M mission in future research, the plume limit value is taken as 0.5 ppm, which corresponds to the desired instrument noise for the mission. Some studies, like Kuhlmann et al. (2019), define the plume based on much smaller threshold values, which make the former larger, with more pixels further from the source. However, as differences become larger further away, these extra plume pixels are not needed, when considering the overarching goal of estimating terrestrial fluxes. This plume definition step is illustrated using Panoply (Figure 4a) and Python (Figure 4b).



(a) Berlin city plume using Panoply.



(b) Berlin city plume after applying threshold mask to matrix of data in Python.

Figure 4: Berlin city plume visualization.

4.5.2 Metrics Definition

This step of the analysis definition consists in setting metrics that quantitatively characterize the plumes (and other signals later studied). The main metric defined and used throughout the rest of the project is the signal size. This measure is defined as the mean CO_2 mole fraction value of all pixels that constitute the city plume. Thus, a model's simulated city plume signal has one signal size value for every timestamp. Furthermore, as one of the goals is to analyse the effect of uncertainties from different signals on the city plume signal, their signal size is also taken as the mean of the city plume pixels. In other words, the same city plume spatial mask at a time t , is applied to the other signals, from the biosphere, background and other anthropogenic sources. Another interesting metric that describes result uncertainty is the plume overlap. It is defined as the fraction of pixels that spatially coincide between models. This metric provides a better understanding of wind variability between WRF configurations. Although due to time limitations from the internship, this final metric could not be utilized in the project, it would improve the overall robustness of the analysis for future research on the topic.

4.5.3 Methodology and Statistical Measures

After having characterized the simulated signals, the last question regarding the analysis setup remains: how to effectively and systematically compare the simulation results? This is done by comparing models two at a time, that differ only in one physical parameterization (PBL scheme, LSM, CP or US scheme), using their signal size over time. Then, statistical measures are defined to quantify and describe the differences in signal size, from model to model, which is ultimately this project's estimation of WRF uncertainty due to transport error. Those measures are the Mean Absolute Difference (MAD), Overpass Mean Absolute Difference (OMAD) and MAX Difference (MAXD). The MAD gives an indication on the general variability between two models, while the OMAD is computed to make sure that variability at overpass remains similar to the rest of the time series. The MAXD gives a better idea on the maximum possible uncertainty associated to the physical parameterization at hand. The process is repeated for all signals, after which the estimation of the uncertainty for each signal can be compared to the city plume signal size, hence giving an idea of the contribution of different CO_2 sources on the uncertainty obtained.

5 Outcome

As mentioned in Section 4.5, a python script is written to convert the WRF output data into intuitive plots illustrating the evolution of the signal size of the different simulated CO₂ signals, from the city plume, biosphere, other anthropogenic sources and background. Those plots represent the final results of the study, and provide the necessary grounds on which to draw conclusions and answer the research questions.

5.1 Model Comparison

The analysis methodology described in Section 4.5 is thus followed for the creation of the plots. The signal size of each signal, for all models, is computed at every time step. Before creating the comparison plots, the signal sizes of a specific signal, for all models, are plotted together to verify that no error occur. However, an initial unexpected behavior was detected in the results. Indeed, the signal size over time, from all sets of models that differed only in the CP were essentially identical. This realization works as the first concrete result from the study, and reduces the remainder of the analysis to only models that differ in the PBL scheme, LSM and US scheme. The models' resultant signal sizes are then plotted in pairs, for every signal, such that the corresponding models vary only in one physical parameterization. The previously defined statistical measures, namely the MAD, OMAD and MAXD are finally computed and serve as the comparison tool between models. The following plots (see Figure 5) show the model to model comparisons produced to evaluate simulated CO₂ mole fraction variability due to physical parameterization. As a large quantity of total plots are available, only the ones that look at the city plume signal are presented, with the results from the rest of the signals summarized in Section 5.2.

The results obtained in Figure 5 can be analysed thus providing insight into the effect of the different physical parameterizations studied on the simulated quantities. All statistical measures used show that the PBL scheme leads to the largest uncertainty in signal size within pairs of models, with an average MAD of 0.077 ppm, closely followed by the LSM, at 0.07 ppm. The variability due to the use or not, of a US scheme is overall smaller than the other options, with a MAD of 0.04 ppm. As later presented in Section 5.2, this pattern of uncertainty between the physical parameterizations occurs for every CO₂ signal. Another interesting result is that the second pair of models (5 and 11) that tests US scheme uncertainty exhibits slightly larger measures of variability than the first pair (models 1 and 9). This behavior is also visible for models differing in LSMs and for other signals, which would indicate that US scheme and LSM based uncertainty increases when the YSU PBL scheme is used (as models 5, 7 and 11) rather than MYNN. Similarly, the third pair of models (9 and 11) that tests PBL scheme uncertainty exhibits slightly larger measures of variability than the other two pair, which suggests that the use of a US scheme increases variability due to PBL schemes. Furthermore, the values of OMAD obtained show that the uncertainty at time of satellite overpass remains very close to the general MAD, meaning that results outside the overpass period can still be used for such research.

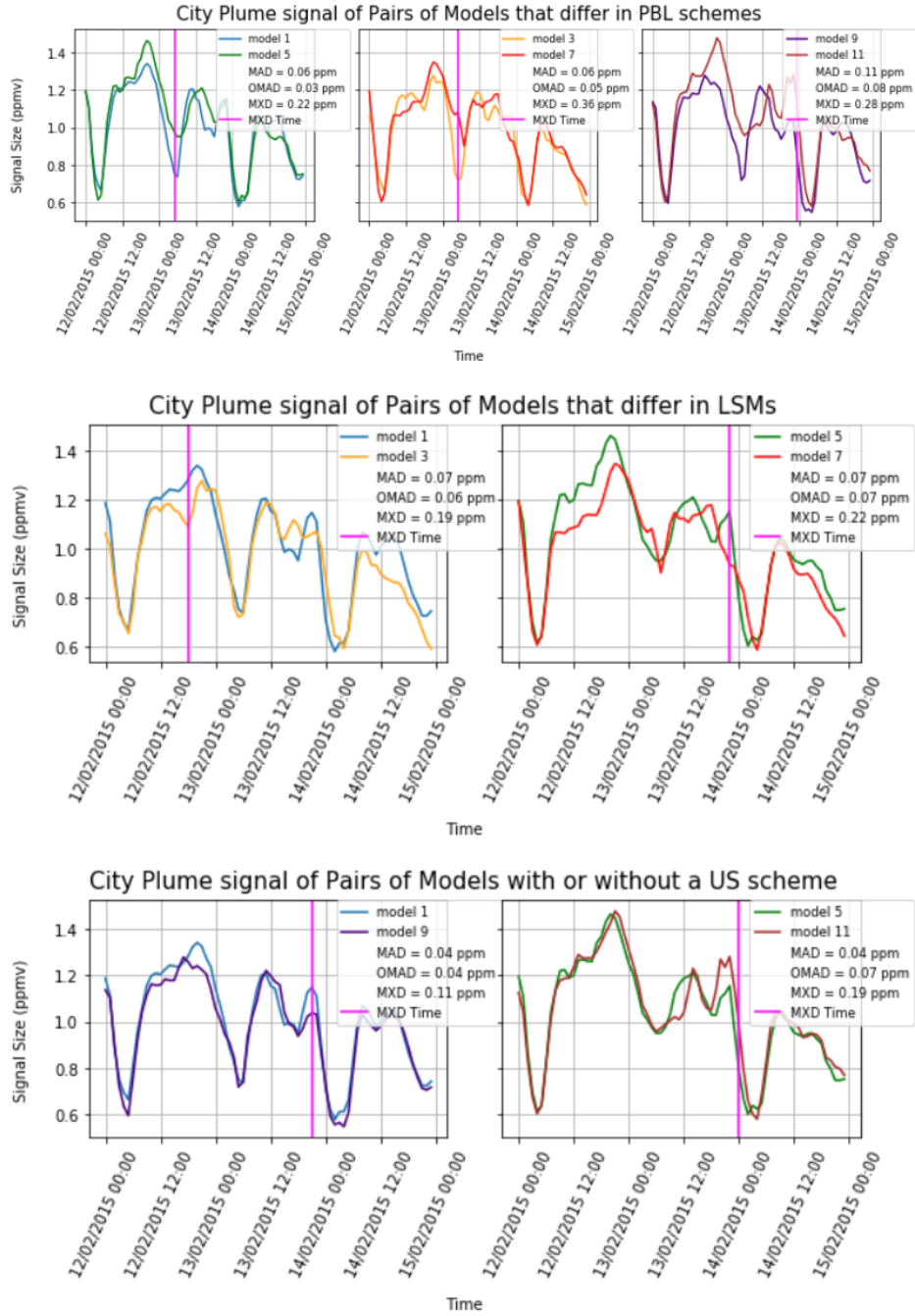


Figure 5: Comparison of models that differ in one physical parameterization scheme, with respect to the city plume signal.

5.2 Results Overview

As previously mentioned, only plots related to the city plume signal were presented. The results produced through Python illustrating the uncertainty in CO₂ signal size due to physical parameterization, of the background, biosphere and other anthropogenic sources signals are shown in Table 3. The table presents how large the variability in CO₂ mole fraction (represented by the average MAD) is relative to the mean city plume signal, of 1 ppm. This table gives an estimation of WRF uncertainty from the physical parameterization, for every signal, as well as of the contribution of the different signals to the uncertainty.

Physics Option Diff Signal	PBL scheme	LSM	Urban Surface scheme	CP
City Plume	7.7%	7%	4%	0%
Other Anthropogenic Sources	15.3%	11%	7%	0%
Biosphere	3%	2.5%	1%	0%
Background	4%	3%	1%	0%
Combined	25%	13.5%	7.5%	0%

Table 3: Overview of results: WRF uncertainty relative to city plume signal size.

The first point to note is that the uncertainty linked to the physics options generally follows the same trend, with the PBL scheme having the largest effect, closely followed by the LSM. The US scheme leads to a lesser uncertainty, while the CP essentially doesn't change the simulated results. A possible reason for this latter point is that, as explained in the domain definition, CP is only used for the parent domain, within which the uncertainty is actually present. However, when studying the nested domain, which does not parameterize convection, the differences between models fade. Finally, the table shows that the signal from other anthropogenic sources produces the largest uncertainty in signal size (up to 15% of the average city plume signal size) followed by the city plume signal. This indicates that total anthropogenic emissions, regrouping sources such as cities, power plants and airports, have the largest contribution to overall uncertainty when simulated in WRF.

6 Conclusions

The goal of this study was firstly to quantify the CO₂ column mole fraction uncertainties that arise from WRF transport model errors, and evaluate the contribution of different signals to the overall uncertainty, with respect to Berlin’s city plume. After getting familiar with the context of the enhanced greenhouse effect and atmospheric transport, along with the tools of the research at hand, the WRF model was setup in accordance to the goals, placing the study region around Berlin and running it at a high resolution. Based on a detailed literature review of sensitivity studies with WRF, an ensemble of models that varied in physical parameterizations was developed. From there, an adapted methodology and metrics were defined to optimize the analysis of the ensemble and produce intuitive results from which the research problematics could be answered. The main conclusions to this study come as follows. The uncertainties in CO₂ mole fractions from different signals due to physical parameterizations were quantified successfully, represented by appropriate statistical measures. For the city plume signal, the PBL scheme produced the largest uncertainty, with a MAD between models of 0.077 ppm, followed by the LSM, at 0.07 ppm, and the US scheme at 0.04 ppm. The CP was found to not produce any visible uncertainty, as only present in the parent domain of simulations. This ranking of physics options related to the variability obtained was also apparent for the other simulated CO₂ signals, and generally agreed with related literature. It was also concluded that the use (or lack thereof) of a US scheme had a smaller impact on result variability, even over a large city like Berlin, than the PBL scheme and LSM. The contributions of different signals to the overall uncertainty in simulated CO₂ mole fractions were also determined. The signal from other anthropogenic sources produced the largest uncertainty in signal size with the MAD being up to 15% of the average city plume signal size, which was followed by the city plume signal itself. Thus, it can be concluded that simulating anthropogenic emissions has a larger contribution to overall WRF uncertainty than natural CO₂ fluxes, much larger in magnitude.

Finally, the research questions at the origin of this internship, related to the quantification of CO₂ column mole fraction uncertainty due to transport error in WRF, were answered successfully. The conclusions drawn from the study provide better understanding on the role of physical parameterizations in the area of West Germany and Berlin. The results obtained show that the uncertainties at hand are far from negligible, and still have an impact on subsequent flux estimations. Thus, further research within the modeling community is required to improve atmospheric transport, and WRF parameterization.

7 Recommendations for Future Work

As mentioned earlier, although the research problematics have been answered and insightful conclusions have been drawn from this project, the work done was somewhat limited by the time constraints of the internship. Thus, some recommendations are given for future continuation and improvements to this research. Firstly, to fully complete this study, a couple of elements should be added. One of the main research points in this area has to do with evaluating model configurations and schemes, by comparing simulated PBL variables to real meteorological data. Another key aspect of the research has to do with the overarching flux estimation goal. Indeed, once the uncertainties are quantified, assessing their effect on terrestrial carbon flux estimations is the end goal. Further recommendations are more to add on to the study of variability in results, with various other WRF options. One of the choices made to fit the time constraints was to vertically average CO₂ column mole fractions which indeed simplified the processing and analysis of data. However, an interesting aspect of this study would be the evolution of the uncertainty at different vertical layers of the PBL. The evolution of the uncertainty could also be evaluated with respect to the distance to the city plume source. This would give an idea of the rate at which the uncertainty increases the further away from the emission source. To increase the completeness of the analysis done in this research, other metrics could be defined and used. One that was described in Section 4.5.2 is the plume overlap, which illustrates the variability of wind directions in the PBL from model to model. Furthermore, this study focused on uncertainty associated with physical parameterizations, which play an important role in it. However, many more WRF settings and options have a critical impact on results variability. One of these interesting "non-physical" parameters is the time of the year. Indeed, as photosynthesis mostly takes place in spring and summer, and more landmass is present on the Northern hemisphere, a seasonal cycle in atmospheric CO₂ concentrations occur. The effect of this natural phenomenon on simulated CO₂ would therefore add diversity to this research. Other settings to be studied and varied include the driving meteorology, nudging, and the spin-up time.

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Age Group	Percentage
18-24	10%
25-34	15%
35-44	20%
45-54	25%
55-64	30%
65-74	35%
75-84	40%
85-94	45%
95-104	50%
105-114	55%
115-124	60%
125-134	65%
135-144	70%
145-154	75%
155-164	80%
165-174	85%
175-184	90%
185-194	95%
195-204	100%



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